

Project Title:

Restrict EC2 Instance Termination for Non-Admin Users Using IAM Policy

Objective:

To prevent non-admin users from terminating EC2 instances while still allowing them to perform other EC2 operations like launch, start, and stop.

Services Used:

- AWS IAM
 - Amazon EC2
-

Project Steps:

1. Created Custom IAM Policy

- Navigated to **IAM → Policies → Create Policy**
- Selected **JSON** option
- Defined a policy to explicitly deny EC2 termination

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PreventEC2Termination",
      "Effect": "Deny",
      "Action": "ec2:TerminateInstances",
      "Resource": "*"
    }
  ]
}
```

- Named the policy:
👉 **Deny-EC2-Termination**
-

2. Created IAM User (Developer)

- Navigated to **IAM → Users → Create User**
 - Created user:
 👉 **developer-user**
 - Enabled **AWS Management Console access**
 - Set custom password
-

3. Attached Permissions to User

Attached the following policies:

- ☒ **AmazonEC2FullAccess** (Allow all EC2 actions)
 - ☐ **Deny-EC2-Termination** (Restrict termination)
-

4. Logged in as Developer User

- Used IAM user login URL
 - Logged in with **developer-user credentials**
-

5. Launched EC2 Instance

- Navigated to **EC2 → Launch Instance**
 - Selected:
 - Amazon Linux AMI
 - t2.micro instance type
 - Successfully launched instance
-

6. Tested Termination Restriction

- Selected the running EC2 instance
- Clicked **Instance State → Terminate**

✓ Result:

- Received authorization error:
 - *"You are not authorized to perform ec2:TerminateInstances"*
 - Explicit deny policy enforced
-

7. Verified Admin Access

- Logged in as Admin/Root user
 - Successfully terminated the same EC2 instance
-

✓ Final Outcome

- Developer users:
 - Can create, start, and stop EC2 instances ✓
 - Cannot terminate EC2 instances ✗
 - Admin users:
 - Full control including termination ✓
-

🔍 Key Concept Used

- **Explicit Deny in IAM overrides Allow permissions**
- Even with full access, denied actions cannot be performed

Real-Time Use Case:

Prevent accidental deletion of:

Production servers

Critical applications

Enforce controlled access in DevOps environments

User created:

The screenshot shows the AWS IAM console for the user 'developer-user'. The left sidebar contains navigation links for Identity and Access Management (IAM), Access Management, and Access reports. The main content area displays the user's summary, including their ARN, creation date, and console access status. Below the summary, the 'Permissions' tab is selected, showing a list of permissions policies attached to the user.

Summary

- ARN: `arn:aws:iam:433038876860:user/developer-user`
- Console access: Enabled without MFA
- Created: March 08, 2026, 09:30 (UTC+05:30)
- Last console sign-in: Today

Permissions policies (3)

Permissions are defined by policies attached to the user directly or through groups.

Policy name	Type	Attached via
AmazonEC2FullAccess	AWS managed	Directly
Deny-EC2-Termination	Customer managed	Directly
IAMUserChangePassword	AWS managed	Directly

Policies attached:

This screenshot is identical to the one above, showing the AWS IAM console for the user 'developer-user'. It highlights the 'Permissions' tab, which lists three policies attached to the user: AmazonEC2FullAccess, Deny-EC2-Termination, and IAMUserChangePassword.

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► **Permissions boundary** (not set)

Final output:

The screenshot displays the AWS Management Console interface. At the top, a red error banner states: "Failed to terminate (delete) an instance: You are not authorized to perform this operation. User: arn:aws:iam::433038876860:user/developer-user is not authorized to perform: ec2:TerminateInstances on resource: arn:aws:ec2:us-east-1:433038876860:instance/i-0036a25d14ba43ab2 with an explicit deny in an identity-based policy: arn:aws:iam::433038876860:policy/Deny-EC2-Termination. Encoded authorization failure message: jDsjPV-mpKsfmx0tWwVv1B0sGCH1TLFV30Dhp33WKT8xEz3VIY7L8gATDALK9vD6aR5qUG-8RHa0J7nxP8gShucW8K21chID9m3y6bAyHM2fXCPNo75cHOLEOrSvHmL9Jkow3ts8j7yHavR5wO-6LlmjVMOY5JHw-RiXoF87Hs7H1xsalcWJS9YI-3vPqEFQMhYs4lTLC_uuUeqcN3hH4Wyc0IALcirbhych501g8C6F9bR1CIO3pHi-8miUz3OpuvkXd5e1KbM2AldgYw9R73IdnD6A44y-C6luhmm35Nd9QO2XK9eY3XyEUI3ml-KYb4JxKajR3s98FmL82wk-HWdIQ9gTwFTZk-3ZyZyQyFOXsqMsa4koPVPCTXcnJWXXcmTBX4WcZFZurknB_4q-kG6se9AIAylyhC5k8pmdINRjV37NSXLDKOFyBBsEoBb2Wb3P97JNiudhBIKL3IFWk2_RSI4szwfkAwwbr34pwOWpweFUE2xXDtb-Ezi1ofTP9h2kshQPK93CETWN3tp-K9oMsihK3fjdCphIXCchvAcDH0JfFvjwVPYMFsOKK-wEslWW0aXzwdOmZ1IN6IQhssMfQ0aDui2He8HgEseB4PNxCcCbBTtyCVlprU5viZaZtCawFWv6SAxc-QNAS50Bw86xvY2BxhclBK2VpYLLp_hxoq_fr_GihwMwGQIUJpJlqgQARUJPzZ-68PFQ_YiBbXmsR3X4ICLBIpQkYZF2aJlHYYPwdU7N_00XXujMcNtZQuOaEjTrIjIS5sRkvsQb3NKL5xNDKaAoz39CCGg-MxVfbx_7noCgBA4xbW_AnHGROxvV0Xc2umzh6fpXN5g4nQyDUS2H9BbGd0h89soTPhcg2u4CDe4CBW1NmVgVfEYbOmFtiNLS4b_3-6bCeEs036al1qmPR0xZO82NI".

Below the error message, the "Instances (1/54)" section is visible. It includes a search bar and a list of instances. The first instance listed is "i-0036a25d14ba43ab2 (email-worker(sqs-p2))". The instance details are shown, including the "Details" tab. The "Instance summary" section is also visible, showing the instance's state and configuration.

The bottom of the screenshot shows the Windows taskbar with the date and time as 5:40 PM on 4/4/2026.